Linear and Discrete Optimization Modelling

UNSW MATH3171 25T3

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Contents

Chapter 1

Introduction to Optimization

1.1 Optimization: An Introduction

Many problems we counter in a large number of area entail trying to achieve the "best" outcome subject to some "restrictions". The mathematical field that encompasses these problems is called Optimization.

An optimization problem consist of three parts: decision variables, object functions and constraints.

(1) **Decision variables**: what can you change or what you wish to determine.

Denote the decision variable $\mathbf{x} \in H$ where H is a finite dimensional space. For example,

- $H = \mathbb{R}^n$ and the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$, the number of variables n.
  - Column vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$;
  - Univariate: $n = 1$;
  - Multivariate: $n = 2, 3, 4, \ldots$ up to millions;
- $H = \mathbb{R}^{p \times n}$ where $\mathbb{R}^{p \times n}$ is the set of $(p \times n)$ matrices, and the decision variable $\mathbf{x} = X \in \mathbb{R}^{p \times n}$ is an $(p \times n)$ matrix.

(2) **Objective functions**: a mathematical function of the variables quantifying the idea of "best".

- $H = \mathbb{R}^n$, the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$ and objective function $f : \mathbb{R}^n \to \mathbb{R}$. For example,
  - Linear: $f(\mathbf{x}) = \mathbf{c}^T\mathbf{x}, \quad \mathbf{c} \in \mathbb{R}^n$ fixed;
  - Affine: $f(\mathbf{x}) = \mathbf{c}^T\mathbf{x} + \gamma, \quad \mathbf{c} \in \mathbf{x}^n, \gamma \in \mathbb{R}$ fixed;
  - Quadratic: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T G\mathbf{x} + \mathbf{c}^T\mathbf{x} + \gamma \quad G \in \mathbb{R}^{n \times n}, \mathbf{c} \in \mathbb{R}^n, \gamma \in \mathbb{R}$ fixed.
- $H = \mathbb{R}^{p \times n}$, the decision variable is an $(p \times n)$ matrix X and objective function $f : \mathbb{R}^{p \times n} \to \mathbb{R}$. For example,
  - $f(X) = \sum_{i=1}^{p} \sum_{j=1}^{n} A_{ij}X_{ij}$, where A is given $(p \times n)$ matrix X;
  - $f(X) = \text{rank}(X)$, where $\text{rank}(X)$ is the rank of a matrix X.

(3) **Constraints**: describe restrictions on the allowable values of variables.

- $H = \mathbb{R}^n$,
  - Equality constraints $g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m_E$ (e.g. binary constraint: $x_i \in \{0, 1\}$);
  - Inequality constraints $g_i(\mathbf{x}) \le 0, \quad i = m_E + 1, \dots, m$;
  - Geometric constraints $\mathbf{x} \in C \subseteq \mathbb{R}^n$;
  - Feasible region $\Omega \in \mathbb{R}^n$

$$\Omega = \left\{ \mathbf{x} \in C : \quad \begin{array}{r} g_i(\mathbf{x}) = 0, i = 1, \dots, m_E; \\ g_i(\mathbf{x}) \le 0, i = m_E + 1, \dots, m \end{array} \right\}$$

- The case for $H = \mathbb{R}^{p \times n}$ is similar, except the constraint functions g_i are functions on $\mathbb{R}^{p \times n}$ and the set C in the geometric constraints is a set in $\mathbb{R}^{p \times n}$. In this case, the feasible region $\Omega \subset \mathbb{R}^{p \times n}$

$$\Omega = \left\{ X \in C \subseteq \mathbb{R}^{p \times n} : \quad \begin{array}{r} g_i(X) = 0, i = 1, \dots, m_E; \\ g_i(X) \le 0, i = m_E + 1, \dots, m \end{array} \right\}$$

Optimization Problem A typical optimization problem looks like

$$\begin{array}{rl} \min\limits_{x \in H} & f(\mathbf{x}) \\ \text{such that} & g_i(\mathbf{x}) = 0, i = 1, \dots, m_E, \\ & g_i(\mathbf{x}) \le 0, i = m_E + 1, \dots, m, \\ & x \in C \subseteq H \end{array}$$

where H is the finite dimensional space.

Linear Programming A linear program is an optimization problem where the underlying space $H = \mathbb{R}^n$, all the functions $f, g_i, i = 1, \dots, m$ are affine functions on $\mathbb{R}^n$ and there is no geometric constraints:

$$\begin{array}{rl} \min\limits_{x \in \mathbb{R}^n} & \mathbf{c}^T \mathbf{x} + \gamma \\ \text{such that} & \mathbf{a}_i^T \mathbf{x} + b_i = 0, i = 1, \dots, m_E, \\ & \mathbf{a}_i^T \mathbf{x} + b_i \le 0, i = m_E + 1, \dots, m. \end{array}$$

Conic Linear Program A conic linear program is an optimization problem where the underlying space H is a general finite dimensional space, all the functions $f, g, i = 1, \dots, m$, are affine functions on H and the set in the geometric constraint is a specific set called convex cone.

Linear Integer Programming A linear integer programming problem is an optimization problem where the underlying space $H = \mathbb{R}^n$, all functions $f, g_i, i = 1, \dots, m$ are affine functions on $\mathbb{R}^n$ and the geometric constraint C is contained in the set of integers:

$$\begin{array}{rl} \min\limits_{x \in \mathbb{R}^n} & f(\mathbf{x}) \\ \text{such that} & g_i(\mathbf{x}) = 0, i = 1, \dots, m_E, \\ & g_i(\mathbf{x}) \le 0, i = m_E + 1, \dots, m, \\ & x \in C \subseteq \mathbb{Z}^n \end{array}$$

Chapter 2

Mathematical Background

2.1 Vectors and Matrices

These notes will skip some preliminaries that you should know from first and second year.

If $\mathbf{x}^T y = 0$, then we say the two vectors are $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Definition. (Vector Norm) A *vector norm* on $\mathbb{R}^n$ is a function $\|\cdot\|$ from $\mathbb{R}^n$ to $\mathbb{R}$ such that

1. $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = 0$.
2. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. (Triangle inequality)
3. $\|\alpha \mathbf{x}\| = |\alpha| \|\mathbf{x}\|$ for all $\alpha \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$.

An $(n \times n)$ matrix A is said to be symmetric if $A = A^T$. We consider eigenvalues in non-ascending order.

Theorem. (Spectral Decomposition Theorem) Let $A \in \mathbb{R}^{n \times n}$ be a symmetric $n \times n$ matrix. Then, there exists an orthogonal matrix Q (that is, $Q^T Q = Q Q^T = I_n$) such that $A = QDQ^T$ where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $Q = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$ where $\mathbf{v}_i$ is an eigenvector of A corresponding to the eigenvalue λ_i.

For a symmetric $(n \times n)$ matrix A and any $\mathbf{x} \neq 0$, we define the so-called Rayleigh quotient as

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\|\mathbf{x}\|_2^2}.$$

Then, we have, for any $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_{\min}(A) \leq R_A(\mathbf{x}) \leq \lambda_{\max}(A).$$

Theorem. (Variational Characterization for Extremal Eigenvalues) For a symmetric $(n \times n)$ matrix A,

$$\lambda_{\max}(A) = \max\{R_A(\mathbf{x}) : \mathbf{x} \neq 0\} = \max_{\mathbf{x} \in \mathbb{R}^n} \{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\|_2 = 1\}$$

and

$$\lambda_{\min}(A) = \min\{R_A(\mathbf{x}) : \mathbf{x} \neq 0\} = \min_{\mathbf{x} \in \mathbb{R}^n} \{\mathbf{x}^T A \mathbf{x} : \|fx\|_2 = 1\}.$$

Definition. (Positive Definite Matrices) A matrix $A \in \mathbb{R}^{n\times n}$ is

- *positive definite* $\iff$ $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0$
- *positive semi-definite* $\iff$ $\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *negative definite* $\iff$ $\mathbf{x}^T A\mathbf{x} \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *indefinite* $\iff$ there exists $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \mathbf{x}^T A\mathbf{x} > 0$ and $\mathbf{y}^T A\mathbf{y} < 0$

A symmetric matrix $A \in \mathbb{R}^{n\times n}$

- has n real eigenvalues $\lambda_i, i = 1, \ldots, n$
- Determinant: $\det(A) = \prod_{i=1}^n \lambda_i$
- Trace: $\operatorname{tr}(A) := \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$

Proposition. (Matrix Definitness) A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is

- *positive definite* $\iff$ $\lambda_i > 0$ for all $i = 1, \ldots, n$
- *positive semi-definite* $\iff$ $\lambda_i \geq 0$ for all $i = 1, \ldots, n$
- *negative definite* $\iff$ $\lambda_i < 0$ for all $i = 1, \ldots, n$
- *negative semi-definite* $\iff$ $\lambda_i \leq 0$ for all $i = 1, \ldots, n$
- *indefinite* $\iff$ there exists $i, j : \lambda_i > 0$ and $\lambda_j < 0$

Theorem. (Checking Positive (Semi-)definitness) For any eigenvalues λ of a symmetric matrix $A \in \mathbb{R}^{n\times n}$, there exists $i_0 \in \{1, \ldots, n\}$ such that

$$\lambda - A_{i_0 i_0} \leq \sum_{j\neq i_0} |A_{i_0} j.$$

Definition. (Diagonally Dominant) A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is called diagonally dominant if

$$|A_{ii}| \geq \sum_{j\neq i} |A_{ij}|, \text{ for each } i \in \{1, \ldots, n\}.$$

In addition, if

$$|A_{ii}| > \sum_{j\neq i} |A_{ij}|, \text{ for each } i \in \{1, \ldots, n\},$$

then we say A is strictly diagonally dominant.

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Suppose that A is diagonally dominant and all of its diagonal elements are non-negative, then any eigenvalue of A must be non-negative, and so, A is positive semi-definite.

Definition. (Matrix Norm) A *matrix norm* on $\mathbb{R}^{p\times n}$ is a function $\|\cdot\|$ from $\mathbb{R}^{p\times n}$ to $\mathbb{R}$ such that

1. $\|X\| \geq 0$ for all $X \in \mathbb{R}^{p\times n}$ and $\|X\| = 0 \iff X = 0_{p\times n}$.
2. $\|X + Y\| \leq \|X\| + \|Y\|$ for all $X, Y \in \mathbb{R}^{p\times n}$. (Triangle inequality)
3. $\|\alpha X\| = |\alpha|\|X\|$ for all $\alpha \in \mathbb{R}, X \in \mathbb{R}^{p\times n}$.

One of the most widely used matrix norm is: **Frobenius norm**: $\|X\|_F = \left(\sum_{i=1}^{p}\sum_{j=1}^{n} X_{ij}^2\right)^{\frac{1}{2}}$.

We can also define a dot product between two matrices $A, B \in \mathbb{R}^{p\times n}$ as

$$A \cdot B = \operatorname{tr}(A^T B).$$

2.2 Local and Global Minima

Let H be a finite dimensional space and let Ω be a set in H. Let $f : H \to \mathbb{R}$ be a function on H and let $\|\cdot\|$ be a norm in H.

Definition. (Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$. The global minimum is $f(\mathbf{x}^*)$.

Definition. (Strict Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *strict global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \Omega, \mathbf{x} \neq \mathbf{x}^*$.

Definition. (Local Minimum) A point $\mathbf{x}^* \in \Omega$ is a *local minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $\|\mathbf{x} - \mathbf{x}^*\| \leq \delta$. Then $f(\mathbf{x}^*)$ is a local minimum.

Definition. (Strict Local Minimum) A point $\mathbf{x}^* \in \Omega$ is a *strict local minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $0 < \|\mathbf{x} - \mathbf{x}^*\| \leq \delta$.

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Definition. (Closed Sets) A set Ω is said to be closed if it contains all the limits of convergent sequences of points in Ω.

Definition. (Bounded Sets) A set Ω is said to be bounded if there exists $R > 0$ such that $\Omega \subseteq \{x : \|x\| \leq R\}$.

Proposition. (Existence of Global Solution: Type I) Let H be a finite dimensional space and let Ω be a closed and bounded set in H and let f be continuous on Ω. Then the global minima of $f$over Ω exist.

Proposition. (Existence of Global Solution: Type II) Let H be a finite dimensional space and let $\|\cdot\|$ be a norm in H. Let Ω be a closed set in H and let f be continuous on Ω. Suppose that f is coercive in the sense that

$$\lim_{\|x\| \to \infty} f(x) = +\infty.$$

Then the global minima of f over Ω exist.

Relaxation Let H be a finite dimensional space. If $f : H \to \mathbb{R}$ and $\overline{\Omega} \subseteq \Omega$ then

$$\min_{x \in \Omega} f(\mathbf{x}) \leq \min_{\mathbf{x} \in \overline{\Omega}} f(\mathbf{x}).$$

2.3 Convexity for Functions/Sets in Euclidean Spaces

2.3.1 Convex Functions

Definition. A set $C \subseteq \mathbb{R}^n$ is convex if for any $\mathbf{x}, \mathbf{y} \in C$ and any $\lambda \in [0, 1]$ the following is true

$$\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in C.$$

In other words, C is convex means the line segment joining the two points $\mathbf{x} \in C$ and $\mathbf{y} \in C$ is completely contained in the set C.

Basic properties of convexity:

- Intersection of convex sets is still convex: Let $\Omega_1, \Omega_2 \subseteq \mathbb{R}^n$ be convex. Then the intersection $\Omega = \Omega_1 \cap \Omega_2$ is convex.
- Intersection of any finite number of convex sets is convex.
- Union of convex sets may or may not be convex.

Definition. (Convex Functions) Let $\Omega \subseteq \mathbb{R}^n$be a convex set. The function $f : \Omega \to \mathbb{R}$ is said to be convex if for all $\theta \in [0, 1]$ and for all $\mathbf{x}, \mathbf{y} \in \Omega$,

$$f(\theta \mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y}).$$

f is strictly convex if and only if $f(\theta \mathbf{x} + (1 - \theta)\mathbf{y}) < \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y})$.

Definition. (Graident) Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable. The graident $\boldsymbol{\nabla} f : \mathbb{R}^n \to$

$\mathbb{R}^n$ of f at $\mathbf{x}$ is

$$\boldsymbol{\nabla} f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n}. \end{bmatrix}$$

Definition. (Hessian) Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. The Hessian $\boldsymbol{\nabla}^2 : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ of f at $\mathbf{x}$ is

$$\boldsymbol{\nabla}^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n^2} \end{bmatrix}$$

Proposition. Let $\Omega \subseteq \mathbb{R}^n$ be a convex set and $f \in C^2(\Omega)$ (f is twice differentiable on Ω).

- $\boldsymbol{\nabla}^2 f(\mathbf{x})$ positive semi-definite for all $\mathbf{x} \in \Omega$ if and only if f is convex on Ω.
- If $\boldsymbol{\nabla}^2 f(\mathbf{x})$ positive definite for all $\mathbf{x} \in \Omega$, then $f$is strictly convex on Ω.

Operations that preseve convexity of functions: Let Ω be a convex set.

1. Nonnegative scaling: If $f$is a convex function on Ω and $\alpha \geq 0$, then αf is also convex on Ω;
2. Sum: If f_1, f_2 are convex on Ω, then $f_1 + f_2$ is also convex on Ω;
3. Pointwise maximum: If $f_i, i = 1, \ldots, m$ are convex function on Ω, then $\max_{1\leq u\leq m} f_i$ is also convex on Ω.

Proposition. Let I be an index set and let $c_i : \mathbb{R}^n \to \mathbb{R}$ be convex functions, $i \in I$. Then

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : c_i(\mathbf{x}) \leq 0, i \in I\}$$

is a convex set.

Definition. (Convexity in the Space of Matrices) A set $C \subseteq \mathbb{R}^{p\times n}$ is convex if for any $X, Y \in C$ and any $\lambda \in [0, 1]$the following is true

$$\lambda X + (1 - \lambda)Y \in C.$$

Definition. (Convex Functions in Matirx Variable) Let $\Omega \subseteq \mathbb{R}^{p\times n}$ be a convex set. The function $f : \mathbb{R}^{p\times n} \to \mathbb{R}$ is said to be convex if for all $\theta \in [0, 1]$, and for all $X, Y \in \Omega$,

$$f(\theta X + (1 - \theta)Y) \leq \theta f(X) + (1 - \theta)f(Y).$$

Definition. Let H be a finite dimensional space. Let $f : H \to \mathbb{R}$ be a function on H and let Ω be a set in H. A **convex programming problem** has

- a convex feasible region Ω
- a convex objection function $f(\mathbf{x})$

Proposition. For a convex programming problem, any local minimizer of f on Ω is a global minimizer of f on Ω.

Chapter 3

Linear Programs

The standard form of a linear programming problem can be formulated as follows

$$\begin{aligned} \text{Minimise}_{\mathbf{x}\in\mathbb{R}^n} \quad & \mathbf{c}^T\mathbf{x} + \gamma \\ \text{subject to} \quad & \mathbf{a}_i^T\mathbf{x} = b_i, i = 1, \dots, m_E, \\ & \mathbf{x} \geq 0, \end{aligned}$$

where $\mathbf{c}, \mathbf{a}_i \in \mathbb{R}^n, b_i \geq 0, i = 1, \dots, m$ and $\gamma \in \mathbb{R}^n$. Throughout this chapter, we always assume that the feasible region is not empty.

3.1 Geometry of Linear Programming Problems

The geometry of the feasible region:

- A **hyperplane** in $\mathbb{R}^n$ is the set of all points $\mathbf{x}$ that satisfy a linear equation

  $$\mathbf{a}^T\mathbf{x} = b,$$

  for some $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$;

- A (closed) **halfspace** in $\mathbb{R}^n$ is the set of all points $\mathbf{x}$ that satisfy the linear inequality

  $$\mathbf{a}^T\mathbf{x} \leq b,$$

  for some $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

- The intersection of a finite number of halfspaces is called a **polyhedron**.

- A polyhedron that is bounded is called a **polytope**.

Note: (1) Finite intersection of polyhedrons is still a polyhedron. (2) Polyhedron is a convex set and the feasible region of a linear programming problem is a polyhedron. In particular, the feasible region of a linear programming problem is a convex set.

(1) The sum $\lambda_1\mathbf{x}_1 + \cdots + \lambda_k\mathbf{x}_k$ is called a **convex** combination of $\mathbf{x}_1, \dots, \mathbf{x}_k$ if the λ_i are all nonnegative and $\lambda_1 + \cdots + \lambda_k = 1$.

(2) The **convex hull** of a set S is denoted by $\operatorname{co} S$ and is defined by

$$\operatorname{co}(S) = \left\{ \lambda_1 \mathbf{x}_1 + \cdots + \lambda_k \mathbf{x}_k : \mathbf{x}_1, \ldots, \mathbf{x}_k \in S, \lambda_i \in [0,1], \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

That is, convex hull of S is the set of all convex combinations of elements of S.

Let P be a convex set in $\mathbb{R}^n$. A vector $\mathbf{x} \in P$ is a **extreme point** of P if whenever $\mathbf{x} = \lambda \mathbf{y} + (1-\lambda)\mathbf{z}$ for some points $\mathbf{y}, \mathbf{z} \in P$ and $\lambda \in [0,1]$ then $\mathbf{y} = \mathbf{z} = \mathbf{x}$.
In other words, the vector $\mathbf{x}$ cannot be written as non-trivial convex combination of two different vectors $\mathbf{y}$ and $\mathbf{z}$. Roughly speaking, $\mathbf{x}$ is an extreme point of P if $\mathbf{x}$ does not lie in any open line segment joining two points of P.

Theorem. Let P be a bounded polyhedron (or P is a polytope) in $\mathbb{R}^n$. Then, P has only finitely many extreme points, and P is a convex hull of its extreme points.

Theorem. Consider a linear programming problem P in standard form. Let Ω be the feasible region of P, that is,

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m \text{ and } \mathbf{x} \geq 0\}.$$

Suppose that Ω is bounded. Then, there must exist an optimal solution of P which is an extreme point of the feasible region Ω.

Theorem. Let Ω be the feasible region of a linear programming problem in standard formulation. Suppose that $m \leq n$ and the collection of vectors $\{\mathbf{a}_1, \ldots, \mathbf{a}_m\}$ are linearly independent. Denote $A = \begin{pmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{pmatrix} \in \mathbb{R}^{m \times n}$ and let $A_j \in \mathbb{R}^m, j = 1, \ldots, n$ be the jth column of A. Then $\mathbf{x}^* = (x_1^*, \ldots, x_n^*) \in \mathbb{R}^n$ is an extreme point of Ω if and only if the following two conditions hold

(1) $\mathbf{a}_i^T \mathbf{x}^8 = b_i, i = 1, \ldots, m, \mathbf{x}^* \geq 0$;

(2) there exists indices $1 \leq j_1, \ldots, j_m \leq n$ such that the column vector $\{A_{j_1}, \ldots, A_{j_m}\}$ is linearly independent and $x_j^* = 0$ for all $j \notin \{j_1, \ldots, j_m\}$.

3.2 Simplex Method

Definition. Consider a linear programming problem (LP) in standard form and denote its feasible region by Ω. A point $\mathbf{x}^*$ is called a **basic feasible solution** of (LP) if $\mathbf{x}^*$ is an extreme point of the feasible region Ω.

Definition. Let $\mathbf{x}$ be a basic feasible solution of the linear programming problem in standard form. The variables with zero values are called **nonbasic variables** associated with the point $\mathbf{x}$. The remaining variables are called **basic variables**.

Summary of the basic simplex method

1. Form the initial simplex tableau and find out an initial basic feasible solution. If this is feasible then proceed to the next step.

2. Determine the largest positive entry associated with the nonbasic variables in the cost row. This will indicate the entering basic variable. If no entries in the firsr row are positive the STOP, this is the optimal solution.

3. Determine the outgoing basic variable by finding the minimum ratio of the b entry over the (positive) entry in the column of the incoming variable. The entry in this row and column is the pivot. Use Gaussian elimination to make all other entries in this pivot column zero and divide through the pivot row by the pivot so that the pivot changes to 1. Go back to item 2.

We now introduce the two phase simplex method. In this method, we first find an initial basic feasible solution in phase 1. Then, we perform the basic simplex method to find the desired optimal solution.

1. Convert the problem to the correct form yb adding slack, surplus and artificial variables.
2. Consider an artificial problem by minimizing the sum of all artificial variables.
3. Eliminate the entries in the first row of the tableau corresponding to the basic artificial variables.
4. Use the basic simplex method on this problem. This will give an initial basic feasible solution (if it exists) for phase 2. If the solution in phase 1 has any of the artificial variables being nonzero then a feasible solution does not exist for this problem.
5. Create the initial tableau for phase 2 by using the final tableau in phase 1 and
   (a) removing all columns associated with artificial variables
   (b) replacing the cost row with the correct one for this new problem
6. Change the entries in the cost row corresponding to the basic variables to zero using the usual updating mechanism and then continue with the basic simplex method.

3.3 Duality Theory and Sensitivity Analysis

Consider the linear programming problem in standard form:

$$
\begin{aligned}
(P) \quad \text{Minimise}_{\mathbf{x}\in\mathbb{R}^n} \quad & \mathbf{c}^T\mathbf{x} + \gamma \\
\text{subject to} \quad & \mathbf{a}_i^T\mathbf{x} = b_i, i = 1, \ldots, m_E, \\
& \mathbf{x} \geq 0,
\end{aligned}
$$

Let us call it the primal problem. The dual problem will be defined as

$$
\begin{aligned}
(D) \quad \text{Maximize}_{\mathbf{y}\in\mathbb{R}^m} \quad & \gamma + \sum_{i=1}^{m} y_i b_i \\
\text{subject to} \quad & \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i \geq \mathbf{0} \\
& \mathbf{y} \text{ unrestricted}
\end{aligned}
$$

Theorem. (Weak Duality) Let $\mathbf{x}$ be feasible for (P) and $\mathbf{y}$ be feasible for (D), then

$$\mathbf{c}^T\mathbf{x} \geq \mathbf{b}^T\mathbf{y} = \sum_{i=1}^{m} y_i b_i.$$

In particular, the objective function of the dual problem is a lower bound of the objective function for the primal problem.

Theorem. (Strong Duality) The following statements hold:

- If either problem (P) or problem (D) has an optimal solution then so does the other and their objective values are equal.
- If either problem has an unbounded objective value then the other problem has no feasible solution.

Theorem. (Complementary Slackness Condition) Let $\mathbf{x}^*$ be optimal for (P) and $\mathbf{y}^*$ optimal for (D), then the following conditions hold

$$x_j^*(\mathbf{c} - \sum_{i=1}^{m} y_i^* \mathbf{a}_i)_j = 0, \quad j = 1, \ldots, n.$$

Theorem. (Optimality Conditions) Let $\mathbf{x}^*$ be feasible for a primal linear programming problem (P) and let $\mathbf{y}^*$ be feasible for its dual problem (D). Then, $\mathbf{x}^*$ and $\mathbf{y}^*$ are optimal solutions for (P) and (D) respectively if and only if $(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^*, \mathbf{y}^*)$ solves the following equations

$$\begin{aligned} \mathbf{a}_i^T \mathbf{x} &= b_i, i = 1, \ldots, m, \quad x \geq 0 \\ \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i &\geq 0 \\ \mathbf{c}^T \mathbf{x} &= \sum_{i=1}^{m} y_i b_i. \end{aligned}$$

3.4 Further Development of Linear Programs

3.4.1 Interior Point Methods

The basic idea of this method, is moving towards the optimum via points in the (relative)-interior of the feasible region of the linear program.

Standing Assumptions: We suppose that the feasible regions

$$F_P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \mathbf{x} \geq 0\}$$

and

$$F_D = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i \geq 0\}$$

both satisfy the (relative) interior point condition, in the sense that

$$\begin{aligned} F_P^+ &= \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \mathbf{x} > 0\} \neq \emptyset \text{ and} \\ F_D^+ &= \{\mathbf{y} \in \mathbb{R}^m : \mathbf{c} - \sum_{i=1}^{m} y_i \mathbf{a}_i > 0\} \neq \emptyset, \end{aligned}$$

where (P) and (D) are the primal and dual linear programs in standard form. We also assume that $\{\mathbf{a}_1, \ldots, \mathbf{a}_m\}$ is linearly independent.

We then construct the μ-perturbed optimality conditions, $\mu > 0$, as follows

$$\begin{cases} \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, & \mathbf{x} \geq 0 \\ \mathbf{w} = \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i, & \mathbf{w} \geq 0 \\ w_j x_j = \mu, & j = 1, \ldots, n. \end{cases}$$

For each $\mu > 0$, let $(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{w}(\mu))$ be a solution of the above system. Then, let

$$(\mathbf{x}(\mu), \mathbf{y}(\mu), \mathbf{w}(\mu)) \to (\mathbf{x}, \mathbf{y}, \mathbf{w})$$

as $\mu \to 0$. Then $(\mathbf{x}, \mathbf{y}, \mathbf{w})$ is a solution for the system above and so $\mathbf{x}$ and $\mathbf{y}$ are solutions for (P) and (D) respectively.

The sets $\{\mathbf{x}(\mu) : \mu > 0\}$ and $\{\mathbf{y}(\mu) : \mu > 0\}$ are called the **central path** of (P) and (D) respectively, where the parameter $\mu > 0$ is called the **central path parameter**.

General Interior Point Method (Outline)

- Start with an initial point $(x_0, y_0) \in F_P^+ \times F_D^+$. Let $\mu > 0$ and $\gamma \in (0, 1)$.
- Solve the following perturbed optimality system

$$\begin{cases} \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, & \mathbf{x} \geq 0 \\ \mathbf{w} = \mathbf{c} - \sum_{i=1}^m y_i \mathbf{a}_i, & \mathbf{w} \geq 0 \\ w_j x_j = \mu, & j = 1, \ldots, n. \end{cases}$$

- Replace μ by $\gamma\mu$.

3.4.2 Lagrange Multiplier Method and Necessary Optimality Conditions

Consider an optimization problem with equality constraints

$$(P_E) \text{Minimize}_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$
$$\text{Subject to} g_i(\mathbf{x}) = 0, i = 1, \ldots, m.$$

The Lagrangian function for (P_E) is

$$L(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \sum_{i=1}^m y_i g_i(\mathbf{x}).$$

We say the Lagrange multiplier necessary optimality conditions for problem (P_E) holds at $\mathbf{x}$ if

(1) $f(\mathbf{x}) < +\infty$;

(2) $g_i(\mathbf{x}) = 0, i = 1, \ldots, m$;

(3) there exist multipliers $y_i \in \mathbb{R}, i = 1, \ldots, m$ such that

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \mathbf{y}) = \nabla f(\mathbf{x}) + \sum_{i=1}^m y_i \nabla g_i(\mathbf{x}) = \mathbf{0}.$$

We say a point $\mathbf{x}$ is a regular point for (P_E) if $\{\nabla g_i(\mathbf{x}) : 1 \leq i \leq m\}$ is linearly independent.

Proposition. (Lagrange Multiplier Necessary Optimality Conditions) For problem (P_E), let $\mathbf{x}^*$ be a local minimizer which is a regular point. Suppose that $f, g_i, i = 1, \ldots, m$ are continuously differentiable functions around $\mathbf{x}^*$. Then Lagrange multipler optimality conditions for problem (P_E) holds at $\mathbf{x}^*$, that is,

(1) $f(\mathbf{x}^*) < +\infty$;

(2) $g_i(\mathbf{x}^*) = 0, i = 1, \ldots, m$;

(3) there exist multipliers $y_i^* \in \mathbb{R}, i = 1, \ldots, m$ such that

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \mathbf{y}^*) = \nabla f(\mathbf{x}^*) + \sum_{i=1}^{m} y_i^* \nabla g_i(\mathbf{x}^*) = \mathbf{0}.$$

3.5 Applications of Linear Programs

- Transportation problems, involving the transport of goods between various locations.
- Zero sum game theory, in particular, theory of two player zero sum games where each participant's gains or losses are exactly balanced by those of the other participants.
  - Rock-Paper-Scissors
  -

    Theorem. For any 2 person zero sum game, let P be the payoff matrix for this game. Then, we have

    $$V_1^* = V_2^* = V^* \quad \text{(called the minimax value of the game)}.$$

    where

    $$V_1^* = \max_{\mathbf{x} \in S_n} \min_{\mathbf{y} \in S_m} \mathbf{y}^T P \mathbf{x}$$
    $$V_2^* = \min_{\mathbf{y} \in S_m} max_{\mathbf{x} \in S_n} \mathbf{y}^T P \mathbf{x}$$
- Radiotherapy treatment planning problem.

Chapter 4

Network Problems

Network representations of problems appear in many areas. There are electrical networks, communications networks, road networks, networks that represent project planning, networks that represent relationships between objects and quite a few more.

Definition. (Network) A network contains two types of object: *nodes* and *arcs* which join the nodes. The nodes are represented by circles and the arcs by lines (in an undirected network) or arrows (in a directed network). For each arc, there is a number associated with it which is called the length of the arc. Arcs that start and finish at the same node (loops) are not allowed.

Numerically, we can recognise a directed (or undirected) network G with n nodes (labelled as $1, 2, \ldots, n$) via an associated $(n \times n)$ matrix $A = (A_{ij})_{1\le i\le n, 1\le j\le n}$ formed by

$$A_{ij} = \begin{cases} w_{ij}, & \text{if there is an arc from node } i \text{ to node } j \text{ and } w_{ij} \text{ is the corresponding arc length;} \\ 0, & \text{if there is no arc from node } i \text{ to node } j. \end{cases}$$

provided that $w_{ij} \neq 0$ for all i, j.

4.1 The Shortest Path Problem

Definition. (Paths and Cycles - Undirected Network Case)

- A **path** in a undirected network is a sequence of connected, distinct arcs. The length of the path is the sum of the length of the arcs traversed.
- A **cycle** in a undirected network is a path where the starting node and end node are the same.

Definition. (Paths and Cycles - Directed Network Case)

- A **path** in a directed network is a sequence of connected, distinct arcs where the arcs must be traversed in the direction of the arrows, so that the tip of one arrow must join the tail of the next arrow. The length of the path is the sum of the length of the arcs traversed.
- A **directed cycle** is a directed path where the starting node and ending node are the same.

Let the length of the arc from node i to node j be denoted by a_{ij}. If there is no arc connect from node i

to node j, let $a_{ij} = \infty$.

Dijkstra's Method (directed network)

1. Initialisation: Set $u_1 = 0$. This is the permanent label for node 1. For each node j that has an arc connecting it to node 1 set $u_j = a_{1j}$. For all other nodes set $u_j = \infty$. There are tentative labels and these nodes are assigned to the set T of tentatively labelled nodes. The set of permanent labels is $P = \{1\}$. Set iteration number $i = 0$.

2. Update the label: Find $k \in T$ where $u_k = \min_{j \in T} u_j$. Set $T = T \setminus \{k\}$ and $P = P \cup \{k\}$. If $k = n$ stop, the shortest path from node 1 to node n has been found with length u_n.

3. Update the cost: Set $u_j = \min\{u_j, u_k + a_{kj}\}$ for all $j \in T$. Update iteration number i as $i + 1$. GO back to the previous step (that is, the step of updating the label).

For an undirected network, we convert it as a directed network and apply the previous Dijkstra algorithm. In implementing the Dijkstra algorithm for undirected network, we proceeds at the same line as for directed network. The only difference is to ignore the direction of the arcs between the two nodes.

4.2 Network Flow Problem

Another important problem for networks is the network flow problem. For this kind of problem we wish to determine a collection of paths from a given starting point (called the source) to a given destination node (called the sink) whereby the amount of material sent from the starting point to the destination is a maximum.

General formula for a maximum flow problem with a single source and single sink. Denote the set of all nodes by $\mathcal{N}$ and the set of all the arcs by $\mathcal{A}$.

Definition. (Flow; Capacity; Source; Sink)

- We denote the **flow** in an arc $(i, j) \in \mathcal{A}$ by x_{ij} and this represents the amount of material going from node i to node j along that arc.
- The **capacity** c_{ij} of an arc $(i, j) \in \mathcal{A}$, where node $i \in \mathcal{N}$ is at the tail of the arc and node $j \in \mathcal{N}$ is at the tip of the arc, is the maximum amount of goods that can be sent through that arc in that direction. We will assume that the minimum amount that can be sent through is zero.
- A **source** is a node where the flow out of that node exceeds the flow into the node. A source is often denoted by s.
- A **sink** is a node where the flow into that node exceeds the flow out of the node. A sink is often denoted by t.
- The arc (i, j) is said to be *outgoing* from node i, *incoming* to node j, and *incident* to both i and j. Define

$$I(i) = \{j \in \mathcal{N} : (j, i) \in \mathcal{A}\} \text{ and } O(i) = \{j \in \mathcal{N} : (i, j) \in \mathcal{A}\}.$$

Conservation law of the network: For each node i where i is not a source or the sink, total flow into i equals total flow out of a node. Mathematically,

$$\sum_{j\in I(i)} x_{ji} = \sum_{j\in O(i)} x_{ij} \quad \text{for all } i \in \mathcal{N} \setminus \{s,t\}.$$

Capacity constraints

$$0 \le x_{ij} \le c_{ij} \quad (i,j) \in \mathcal{A}.$$

General Maximum Flow Problems for single source and single sink (which arcs may carry factional units of flow)

$$\begin{aligned} \text{Maximise } & \sum_{i\in I(t)} x_{it} \\ \text{subject to } & \sum_{j\in I(i)} x_{ji} = \sum_{j\in O(i)} x_{ij} \quad \text{for all } i \in \mathcal{N} \setminus \{s,t\} \\ & 0 \le x_{ij} \le c_{ij} \quad (i,j) \in \mathcal{A}. \end{aligned}$$

Theorem. Consider a linear programming problem maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \le b, \mathbf{x} \ge 0\}$ with an integer vector b such that the optimal value is finite. If the matrix A is a totally unimodular matrix in the sense that every square submatrix of A has determinant $+1, -1$ or 0, then

- All the extreme point of $\{A\mathbf{x} \le b, \mathbf{x} \ge 0\}$ are integral vectors, and so, this linear programming problem has an integral optimal solution.
- Moreover, the integral solution of maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \le b, \mathbf{x} \ge 0\}$ is also a solution of maximize$\{\mathbf{c}^T\mathbf{x} : A\mathbf{x} \le b, \mathbf{x} \ge 0, \mathbf{x}$ is an integral vector$\}$.

Chapter 5

Conic Linear Programs

5.1 Convex Cones and Conic Programs

5.1.1 Inner Product Spaces

Let H be a finite dimensional space. We say a mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}$ is an inner product in H if

(1) $\langle x+y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$;

(2) $\langle x, y\rangle = \langle y, x\rangle$;

(3) $\langle \alpha x, y\rangle = \alpha\langle x, y\rangle$, for $\alpha \in \mathbb{R}$;

(4) $\langle x, x\rangle \geq 0$ for all $x \in H$, and $\langle x, x\rangle = 0$ if and only if $x = 0$.

A finite dimensional space H together with an inner product on it is called a finite dimensional inner product space. Examples:

- $H = \mathbb{R}^n$ where $\langle x, y\rangle = x^T y$ for all $x, y \in \mathbb{R}^n$;
- $H = \mathbb{R}^{p\times n}$ where $\langle X, Y\rangle = \mathrm{tr}\big(X^T Y\big)$ for all $X, Y \in \mathbb{R}^{p\times n}$;
- $H = S^n$ where $\langle X, Y\rangle = tr(XY)$ for all $X, Y \in S^n$.

Definition. (Cone) A set K in a finite dimensional inner product space is called a cone if, for all $x \in K$ and $\alpha \geq 0$, $\alpha x \in K$. In other words, K is a cone if, for any point $x \in K$, the ray

$$\mathbb{R}_+ x = \{\alpha x : \alpha \geq 0\}$$

is a subset of K.

Basic examples of cones:

1. Nonnegative cone: $K = \mathbb{R}^n_+$;
2. Second-order cone:

$$K = \mathrm{SOC}^n := \{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : \sqrt{x_2^2 + \cdots + x_n^2} \leq x_1\};$$

3. Positive semi-definite cone:

$$K = \mathbb{S}^n_+ := \{X \in \mathbb{R}^{n\times n} : X \text{ is a symmetric positive semi-definite matrix}\};$$

4. A simple two dimensional cone:

$$K = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 x_2 \geq 0\}.$$

Definition. (Convex Cones) A convex cone is a cone that is also a convex set.

Theorem. (Alterntive Definition of a Convex Cone) A cone $K \subseteq \mathbb{R}^n$ is convex if and only if

$$x + y \in K \quad \text{for all } x, y \in K.$$

Proposition. The positive semi-definite cone S^n_+ is a convex cone.

Theorem. Let C_i be convex cones, $i = 1, \ldots, p$. Then, the following statements hold:

1. $\bigcap_{i=1}^p C_i$is a convex cone if $H_1 = H_2 = \cdots H_p = H$ and $C_i \subseteq H$.
2. $\prod_{i=1}^p C_i := C_1 \times C_2 \times \cdots C_p$ is also a convex cone.
3. The Minikowski sum

$$C_1 + C_2 + \cdots + C_p = \{x : x = \sum_{i=1}^p x_i, x_i \in C_i\}$$

is also a convex cone if $H_1 = H_2 = \cdots = H_p = H$.

Note: For two convex cones, $C_1 \cup C_2$ need not be convex!

Definition. (Dual Cone) Let K be a convex cone in an inner product space H. The (positive) dual cone is defined as K^* which is defined by

$$K^* = \{x \in H : \langle z, x\rangle \geq 0 \text{ for all } z \in K\}.$$

In the literature, there is also a notation called polar cone of K is denoted as K° and is defined by

$$K^\circ = -K^* = \{x \in H : \langle z, x\rangle \leq 0 \text{ for all } z \in K\}.$$

Proposition. The positive dual cone of positive semi-definite cones S^n_+ is itself (equivalently, the polar cone of positive semi-definite cone S^n_+ is $-S^n_+$).

Definition. We say a convex cone K is self-dual if $K^* = K$. The nonnegative cone, the second order cone and the positive semidefinite cone are all self-dual convex cones.

Let H be a finite dimensional inner produce space and let $K \subseteq H$ be a convex cone in H. A conic programming problem has the form

$$\begin{aligned} \text{Minimise}_{x\in H} \quad & \langle c, x\rangle \\ \text{subject to} \quad & \langle a_i, x\rangle = b_i, i = 1, \ldots, m, \\ & x \in K, \end{aligned}$$

where c and $a_i, i = 1, \ldots, m$ are elements of an inner product space $H, b_i, i = 1, \ldots, m$ are real numbers and K is a closed convex cone in H. The program is a convex program in which the only non-linearities appear in the cone constraint $x \in K$.

Some special cases of conic programming problems:

- For $H = \mathbb{R}^n, \langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$ and $K = \mathbb{R}^n_+$, then the associated conic programming problem is just the **linear programming problem**.
- For $H = \mathbb{R}^n, \langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^T \mathbf{v}$ and $K = \prod_{j=1}^p \mathrm{SOC}^{n_j}$ with $\sum_{j=1}^n n_j = n$, then the associated conic programming problem

$$\begin{aligned} \text{Minimise}_{x \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{subject to} \quad & \mathbf{a}_i^T \mathbf{x} = b_i, i = 1, \ldots, m, \\ & \mathbf{x} \in \prod_{j=1}^p \mathrm{SOC}^{n_j}, \end{aligned}$$

  is called a primal form (or conic form) of **second-order cone programming problem**. For $H = S^n, \langle A, B \rangle := \mathrm{tr}(AB)$ and $K = S^n_+$, then the associated conic programming problem

$$\begin{aligned} \text{Minimise}_{x \in S^n} \quad & \mathrm{tr}(CX) \\ \text{subject to} \quad & \mathrm{tr}(A_i X) = b_i, i = 1, \ldots, m, \\ & X \in S^n_+ (\text{or } X \succeq 0), \end{aligned}$$

  is called a primal form (or conic form) of **semi-definite programming problem**.

5.2 Dual Problems of Conic Program and Duality Theory

Chapter 6

Solutions to Tutorial Problems

Note the problems are not available here. Please see course resources for reference.

6.1 Optimization Model Formulations

6.2 Mathematical Background

6.3 Geometry of Linear Programs and Simplex Methods

6.4 Duality and Sensitivity Analysis for Linear Programs

6.5 Interior Point Method for Linear Programs and Applications for Linear Programs

6.6 Network Problems

6.7 Conic Programs

6.8 Integer Programs